DNS — The Internet's Directory Service

Introduction

The Domain Name System (DNS) is a fundamental component of the Internet, providing a mapping between human-friendly hostnames and machine-usable IP addresses. Without DNS, accessing online resources would require memorizing long, numerical IP addresses.

DNS acts as:

- A **distributed hierarchical database** implemented via DNS servers.
- An **application-layer protocol** (running over UDP, port 53) that allows hosts to query this database.

Identifiers for Humans and Machines

Humans and computers use different methods of identification:

- Humans prefer mnemonic names (e.g., birth certificate names, hostnames like `www.google.com`).
- Machines prefer structured identifiers (e.g., Social Security Numbers, IP addresses).

Hostnames

Examples: `www.facebook.com`, `gaia.cs.umass.edu`. They are memorable but not efficient for routers.

IP Addresses

An IP address consists of four bytes (IPv4) or sixteen bytes (IPv6). Example: `121.7.106.83`. It is hierarchical: scanning left-to-right provides increasingly specific information about network location.

Services Provided by DNS

Hostname to IP Address Translation

1. User enters URL (e.g., `www.someschool.edu/index.html`).
2. Browser extracts the hostname and invokes DNS client.
3. DNS client queries DNS server.
4. DNS server replies with the IP address.
5. Browser establishes TCP connection to target IP (port 80 for HTTP).

Host Aliasing

A host can have a canonical hostname and several aliases. Example:

- Canonical: `relay1.west-coast.enterprise.com`
- Aliases: `enterprise.com`, `www.enterprise.com`

Mail Server Aliasing

DNS MX (Mail Exchange) records allow mnemonic email addresses. Example:

- User address: `bob@yahoo.com`
- Canonical mail server: `relay1.west-coast.yahoo.com`

Load Distribution

- Multiple replicated servers with distinct IPs.
- DNS returns set of IPs, rotating order (Round-Robin DNS).
- Enhances performance and reliability.

DNS Architecture

Client-Server Paradigm

DNS runs between clients and servers, using UDP (port 53). Unlike HTTP or SMTP, DNS is not a user-facing application but a **core Internet service**.

Hierarchical Structure

DNS servers are organized hierarchically:

1. Root servers.
2. Top-Level Domain (TLD) servers (e.g., `.com`, `.edu`, `.fr`).
3. Authoritative servers for organizations.
4. Local resolvers (typically at ISPs or institutions).

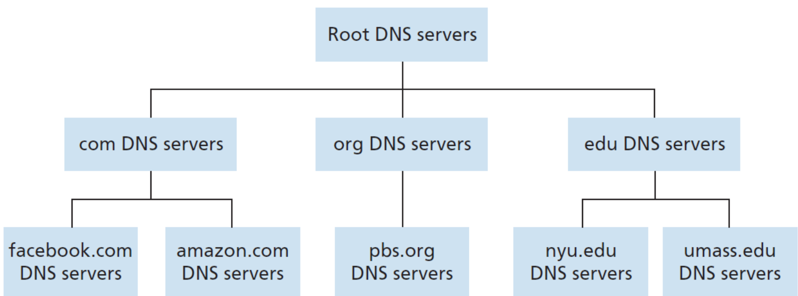

Figure 1: DNS Hierarchy

Performance Considerations

Caching

To reduce delays and traffic, DNS responses are cached with an associated Time-To-Live (TTL).

Delays

DNS lookups can introduce noticeable delays, but caching significantly reduces repeated queries.

DNS Records

DNS stores information as **Resource Records (RRs)** of the form:

$$(\text{Name}, \text{Value}, \text{Type}, \text{TTL})$$

Common record types:

Type	**Purpose**
A	Hostname to IPv4 address
AAAA	Hostname to IPv6 address
MX	Mail server mapping
CNAME	Canonical name (aliasing)
NS	Authoritative name servers

Modern Applications of DNS

- Web browsing (HTTP/HTTPS)
- Email routing (SMTP, IMAP, POP3)
- Content Delivery Networks (CDNs) like Akamai
- Load balancing for large-scale services

Conclusion

DNS is a cornerstone of Internet usability. Its distributed and hierarchical nature ensures scalability, redundancy, and efficiency. Beyond name resolution, DNS provides aliasing, email routing, and load balancing, making it indispensable for the modern Internet.

References

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- Mockapetris, P. (1988, 2005) – DNS Design and Evolution.
- Albitz, P. and Liu, C. (1993) – *DNS and BIND*.
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